Stage 1 for

AI System Design PROJECT
Umm Al Qura University

Project Title: Smart Parking detection with proximity recommendations

Team Members:

Student Name	Student ID	Email	Role
Noran Abdullah Aljodi	445005178	S445005178@uqu.edu.sa	Leader
Reema Salem Alharbi	445000219	s445000219@uqu.edu.sa	Member
Enas Abdullah Saud	44512190	s44512190@uqu.edu.sa	Member
Asail Hamdan Al-Osaimi	445000051	s445000051@uqu.edu.s	Member
Rema Saed Althaqfi	445005754	s445005754@uqu.edu.sa	Member

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1. Problem Definition & Planning

1.1 Problem Statement and Background

Some people face difficulty finding a parking space in the shortest possible time, especially in crowded places such as malls and hospitals. Therefore, there is a need for an automated system capable of detecting the status of the nearest available parking space and enhancing the user experience.

Parking management has become a major challenge in crowded environments. Many drivers spend a long time searching for an available parking space, especially in places that experience heavy congestion such as universities, hospitals, and shopping malls. This not only wastes time but also contributes to increased traffic congestion and fuel consumption.

With the significant advancement in computer vision technologies, it has become possible to develop intelligent systems capable of analyzing images and automatically detecting the status of parking spaces. When these systems are used, they help drivers quickly identify vacant parking spots and guide them to the nearest available space, which improves efficiency and reduces time as well as unnecessary vehicle movement within parking areas.

Therefore, developing a smart parking detection system based on computer vision techniques can provide a practical and applicable solution to improve parking management in real world environments.

1.2 Target User Group

The main users of this system are drivers who are looking for available parking spaces in crowded places such as universities, shopping malls, hospitals, and public parking areas. Many drivers spend a long time searching for empty parking spaces, especially during busy times. This causes delays, traffic congestion, and extra fuel consumption.

The proposed system helps drivers by automatically detecting which parking spaces are available or occupied using computer vision. The system then identifies the nearest available parking space, which helps reduce the time needed to find parking and makes the parking process easier and faster.

Other users of the system may include parking managers or building administrators. They can benefit from understanding how parking spaces are used and how often they are occupied. This information can help improve parking organization and management.

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1.3 Success Metrics and KPIs

The performance of the proposed system will be evaluated using a set of standard metrics commonly used in computer vision to ensure accurate and reliable parking space detection. These metrics include accuracy, which measures the overall correctness of the classification results, precision, which evaluates how many of the predicted available parking spaces are actually correct, and recall, which measures the system's ability to detect all available parking spaces. In addition, the F1-score will be used to provide a balanced evaluation between precision and recall, especially in cases where the dataset may be imbalanced.

Furthermore, a set of key performance indicators (KPIs) will be used to assess the effectiveness of the system in real-world scenarios. These include the detection accuracy rate, system response time in processing images and generating recommendations, and the robustness of the model under varying conditions such as lighting changes and occlusion. The system will also be evaluated based on its ability to reduce the time required for drivers to find parking spaces, as well as its efficiency in guiding users to the nearest available parking spot using shortest-path algorithms. Together, these metrics and KPIs provide a comprehensive evaluation that reflects both the technical performance and the practical applicability of the system.

1.4 Feasibility and Initial Risk Assessment

The project is considered technically feasible because it relies on standard computer vision techniques and publicly available parking datasets. Images are preprocessed with resizing, normalization, and enhancement to ensure model compatibility and stable training performance. The dataset contains labeled parking images taken under various lighting and weather conditions, making it appropriate for realistic evaluation. However, several risks may affect the system, such as lighting variations, poor image quality, occlusion, camera angle differences, and dataset imbalance, all of which may reduce detection accuracy or produce incorrect recommendations. Additional challenges, such as limited computational resources, time constraints, and achieving high accuracy, may impact implementation; however, these risks can be mitigated through preprocessing, dataset balancing, and model adjustment techniques to maintain system reliability.

From an ethical perspective, the project utilizes the PKlot dataset hosted on Kaggle. This dataset is free of personal or sensitive information, such as identifiable faces or license plates, which reduces privacy concerns and making the system suitable for academic use. The data is used solely for research and educational purposes, ensuring compliance with ethical guidelines for computer vision projects.

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1.5 Ethical, Legal, Societal Considerations

1. Ethical Considerations

The proposed AI-based parking system must adhere to fundamental ethical principles to ensure responsible and trustworthy deployment. One of the primary concerns is user privacy, particularly when sensors or camera-based technologies are used to detect parking availability. The system should implement data minimization strategies and avoid collecting personally identifiable information (PII) unless strictly necessary, in accordance with the principles established by Saudi Data and Artificial Intelligence Authority (SDAIA) (SDAIA, 2023).
In addition, fairness and non-discrimination must be ensured. The system should provide equal access and service quality to all users without bias. This requires training AI models on diverse and representative datasets to mitigate potential bias and ensure equitable outcomes.
Transparency and accountability are also critical. Users should be clearly informed about how the system operates, how decisions are made, and how their data is collected and used (OECD, 2021).

2. Legal Considerations

From a legal standpoint, the system must comply with data protection laws and regulations in Saudi Arabia, particularly the Personal Data Protection Law (PDPL), which governs the collection, processing, and storage of personal data (SDAIA, 2023).
If the system relies on surveillance technologies such as cameras, it is essential to obtain the necessary approvals from relevant authorities and ensure compliance with local regulations concerning public monitoring and data usage.
Moreover, all external resources, including datasets, APIs, and software libraries, must be properly licensed and referenced to avoid violations of intellectual property rights.

3. Societal Considerations

The proposed system is expected to deliver significant societal benefits, including reducing traffic congestion, minimizing time spent searching for parking, and enhancing overall user experience. These benefits align with the objectives of Saudi Vision 2030, particularly in promoting smart city initiatives and digital transformation.
However, potential challenges must also be considered. One such challenge is over-reliance on automated systems, where users may depend entirely on AI-driven decisions. Additionally, accessibility should be ensured so that the system can be effectively used by individuals with varying levels of technical expertise.

4. Assumptions and Constraints

Assumptions:

- Users have access to smartphones or digital platforms.
- Parking data is updated accurately and in real time.
- Sensors and data sources provide reliable information.

Constraints:

- Limited availability or quality of data.
- Hardware and infrastructure limitations.
- Legal restrictions on data collection (e.g., camera usage).
- Time and resource constraints during development.

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1.6 User-Centered Design Considerations

The proposed smart parking system follows a user-centered design approach by focusing on the real needs of drivers who often face difficulty finding available parking spaces in crowded environments. The system is designed not only to detect whether parking spaces are available or occupied, but also to assist users in identifying the nearest available parking spot. This helps reduce the time and effort required to search for parking and supports faster decision-making in real-world situations.

In this project, the design considers what is most helpful for the user during the parking search process. Knowing only the availability of parking spaces may still require extra time to locate a convenient spot. Therefore, providing both the availability status and the nearest parking recommendation makes the system more practical and useful for the user.

The project focuses on providing a solution that is useful in real situations, not only achieving good model accuracy. The system is evaluated using standard performance metrics, in addition to measures related to identifying the nearest available parking spot. These considerations ensure that the system supports user needs and improves the overall parking experience.

Initial task distribution among team members

Student Name	Initial task
Enas Abdullah Saud	Evaluate technical feasibility, perform initial risk assessment and mitigation planning, and ensure ethical compliance of the dataset.
Reema Salem Alharbi	Define system evaluation metrics and key performance indicators (KPIs)
Rema Saed Althaqfi	Ensure ethical, legal, and societal compliance in the AI parking system, covering privacy, fairness, transparency, accountability, and accessibility.
Noran Abdullah Aljodi	Identify the main users of the system, describe their needs, apply user-centered design principles, and explain how the system helps users find parking spaces more easily.
Asail Hamdan Al-Osaimi	Define the Problem and Describe the Background

References

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